Robot–Patient Interaction Code: API Reference

module: rpi_gate | version: 1.0.0 | status: reference

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Index terms

Overview

This reference documents the public surface of the robot-patient interaction gate: the entry points a caller uses to submit interaction code for verification before it is generated or executed. Every call is synchronous and returns a gate verdict.

Endpoints

```
verify(interaction: Interaction) -> Verdict
```

Run the full VVUQ gate over a candidate interaction and return ACCEPT, BLOCK, or ESCALATE.

interaction the candidate robot-patient interaction record

Verdict one of ACCEPT, BLOCK, ESCALATE plus the verification fraction

```
attest(verdict: Verdict, officer: str) -> Record
```

Bind a cleared verdict to a responsible officer and emit a hash-chained audit record.

verdict a prior ACCEPT verdict

officer the attesting officer identifier

Data model

Field	Type	Meaning
result	enum	ACCEPT / BLOCK / ESCALATE
fraction	float	verification fraction; must equal 1.0
gates	list	per-gate agreement and CoV bound

Table 1: Table 1. Verdict fields (modeled on VVUQ-05 Table 1).

Architecture

The gate is a pure function of the interaction record and the bound standards; no hidden state is read at call time, so a verdict is reproducible from its inputs.

Figure placeholder (Images/call-graph.png). The float, caption, and \label mirror the VVUQ-02 figures; replace the box with \includegraphics.

Figure 1: Figure 1. Placeholder call graph from verify() through the ten gates to attest().

Errors

Errors are returned as values, not exceptions.

E_FRACTION the verification fraction is below 1.0; the interaction is BLOCKED

E_CATASTROPHE a hard catastrophe predicate failed at agreement 1.00

E_OFFICER attest() called with an unknown officer identifier

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Cite This Manual

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Data Availability

Supporting records are openly available in the kevinkawchak/cancer-automated and kevinkawchak/physical-ai-oncology-trials GitHub repositories and are archived on Zenodo. All data sets are synthetic or illustrative and reproducible from a deterministic seed; no patient-identifiable information is included.

References

[1] K. Kawchak, “VVUQ Processing Priority over Code Generation: Method and Pipeline.” https://doi.org/10.5281/zenodo.20372501, 2026. https://doi.org/10.5281/zenodo.20372501.

[2] K. Kawchak, “Mobile Pancreatic Cancer Unitree H2 Surgical Humanoid with Priority VVUQ.” https://doi.org/10.5281/zenodo.20421754, 2026. https://doi.org/10.5281/zenodo.20421754.

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